Symplectic isogenies of hyperkähler manifolds and Shafarevich conjecture

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Correspondencies

DEFINITION: A cycle in a complex variety is a subvariety $A \subset M$ with **the multiplicity** (a positive integer) attached to any irreducible component of A.

DEFINITION: Let A, B be irreducible compact complex varieties of dimension n. **A correspondence** is an n-dimensional cycle $Z \subset A \times B$ such that each component of Z projects to A and B surjectively.

REMARK: We should consider a correspondence as a "$p : q$-multivalued map." In particular, **any map or any ramified covering define a correspondence.**

REMARK: Let $Z_1 \subset A \times B$ and $Z_2 \subset B \times C$ be correspondencies, and $\pi : A \times B \times B \times C \to A \times C$. the projection. Consider the subvariety $\pi(Z_1 \circ Z_2 \cap \Delta_B) \subset A \times C$, where $\Delta_B \subset A \times B \times B \times C$ is the set (a, b, b, c) for all $a \in A, b \in B, c \in C$. By definition, $(a, c) \in \pi(Z_1 \circ Z_2 \cap \Delta_B)$ whenever there exists $b \in B$ such that $(a, b) \in Z_1$ and $(b, c) \in Z_2$. Therefore, there are irreducible components of $\pi(Z_1 \circ Z_2 \cap \Delta_B)$ such that their projection to A and C is generically finite. Each component of $Z_1 \circ Z_2$ is counted with multiplicity given by the number of components of $Z_1 \circ Z_2 \cap \Delta_B$ projected to $Z_1 \circ Z_2$.

Composition of correspondencies

DEFINITION: The union of all such components is denoted $Z_1 \circ Z_2$. We call it **the composition of the correspondences**.

REMARK: Clearly, **the composition of correspondences is associative.**

CLAIM: Let $Z \subset A \times B$, and $Z^* \subset B \times A$ be its image under the isomorphism $(a,b) \mapsto (b,a)$. **Then $Z \circ Z^* = pq \operatorname{Id}_A$,** where Id_A is the identity correspondence (that is, diagonal), and p,q the degrees of the projection $Z \to A$ and $Z \to B$.

Proof: Left as an exercise. ∎

Correspondencies and holomorphic forms

REMARK: Every isogeny $Z \subset A \times B$ defines a map $H^{p,0}(A) \to H^{p,0}(B)$ taking a holomorphic form η on A to $(\pi_2)_*(\pi_1^*\eta)$. The pushforward of a differential form is (generally speaking) a current, but **a pushforward of a holomorphic form is a holomorphic form, smooth outside of the exceptional set of the map, and by Hartogs it is holomorphic everywhere.**

CLAIM: Let $Z_1 \subset A{\times}B$ and $Z_2 \subset B{\times}C$ be correspondences, and V_{Z_1}, V_{Z_2}, $V_{Z_1 \circ Z_2}$ be the corresponding maps on holomorphic p-forms, $V_{Z_1}: H^{p,0}(A) \to H^{p,0}(B)$ etc. **Then** $V_{Z_1} \times V_{Z_2} = V_{Z_1 \circ Z_2}$.

Proof: The correspondence $Z_1 \subset A \times B$ takes a p-form η on a disk $U \subset A$ to its preimages in Z_1, pushes each preimages to B and takes the sum. If we understand Z_1 as a multi-valued map, it takes η to the sum of all its images in B. By definition, $Z_1 \circ Z_2$ takes $x \in A$ to all $Z_2(b_i)$, where b_i are all images of a under Z_1. Therefore, $Z_1 \circ Z_2(\eta) = Z_2(\mathbb{Z}_1(\eta))$. ■

Hodge structures

DEFINITION: Let $V_{\mathbb{R}}$ be a real vector space. **A (real) Hodge structure of weight** w on a vector space $V_{\mathbb{C}} = V_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ is a decomposition $V_{\mathbb{C}} = \bigoplus_{p+q=w} V^{p,q}$, satisfying $\overline{V^{p,q}} = V^{q,p}$. It is called **rational (integer) Hodge structure** if one fixes a rational (integer) lattice $V_{\mathbb{Q}}$ such that $V_{\mathbb{R}} = V_{\mathbb{Q}} \otimes \mathbb{R}$. A Hodge structure is equipped with $U(1)$-action, with $u \in U(1)$ acting as u^{p-q} on $V^{p,q}$. **Morphism** of Hodge structures is a rational (integer) map which is $U(1)$-invariant.

REMARK: The Hodge decomposition defines an integer Hodge structure on $H^p(X, \mathbb{Z})$ for any compact Kähler manifold X.

CLAIM: The category of integer Hodge structures of weight 1 **is equivalent to the category of complex tori.**

Proof: For each torus T, $H^1(T)$ is equipped with an integer Hodge structures of weight 1. Conversely, if $V_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{C} = V_{\mathbb{C}} = V^{1,0} \oplus V^{0,1}$, the quotient $T := V^{1,0}/V_Z$ is a complex torus such that $H^1(T) = V$. ■

REMARK: Given an isogeny $Z \subset A \otimes B$, the map $\eta \mapsto (\pi_2)_*(\pi_1^* \eta)$ **defines a morphism of Hodge structures** $H^p(A) \to H^p(B)$.

Isogenies of a torus

DEFINITION: Let T_1, T_2 be compact tori of the same dimension. **An isogeny** between T_1, T_2 is a subtorus $T \subset T_1 \times T_2$ such that its projection to T_1, T_2 is a covering. Two tori T_1, T_2 are called **isogeneous** if there exists an isogeny $T \subset T_1 \times T_2$.

REMARK: Correspondencies form a category, with the composition defines as above. It is not hard to see that **a composition of isogenies is an isogeny.**

PROPOSITION: Two tori T_1, T_2 are isogeneous **if and only if the rational Hodge structure on $H^1(T_1)$ is isomorphic to that on $H^1(T_2)$.**

Mumford-Tate group

DEFINITION: Let V be a Hodge structure over $\mathbb{Q}$, and ρ the corresponding $U(1)$-action. **Mumford-Tate group** (Mumford, 1966; Mumford called it "the Hodge group") is the smallest algebraic group over $\mathbb{Q}$ containing ρ.

THEOREM: Let V be a rational Hodge structure, and $\mathsf{MT}(V)$ its Mumford-Tate group. Consider the tensor algebra of V, $W = T^{\otimes}(V)$ with the Hodge structure (also polarized) induced from V. Let W_h be the space of all ρ-invariant rational lines in $\mathbb{P}W$. **Then $\mathsf{MT}(V)$ coincides with the stabilizer**

$$\mathsf{St}_{GL(V)}(W_h) := \{g \in GL(V) \quad | \quad \forall w \in \mathbb{W}_h, g(w) = w\}.$$

Proof: Follows from the Chevalley's theorem on tensor invariants. ∎

Corollary 1: Let $V_{\mathbb{Q}}$ be a rational Hodge structure, and $W \subset V_{\mathbb{C}}$ a subspace. **Then the following are equivalent.**

(i) W is a Hodge substructure.

(ii) W is Mumford-Tate invariant. ∎

Mumford-Tate group and the $\mathcal{G}al(\mathbb{C}/\mathbb{Q})$-action

DEFINITION: Let $V_\mathbb{Q}$ be a $\mathbb{Q}$-vector space, and $V_\mathbb{C} := V_\mathbb{Q} \otimes_\mathbb{Q} \mathbb{C}$ its complexification, equipped with a natural Galois group $\mathcal{G}al(\mathbb{C}/\mathbb{Q})$ action. We call a complex subspace $T \subset V_\mathbb{C}$ **rational** if $T = T_\mathbb{Q} \otimes_\mathbb{Q} \mathbb{C}$, where $T_Q = V_\mathbb{Q} \cap T$.

REMARK: A complex subspace $W \subset V_\mathbb{C}$ is rational if and only if it is $\mathcal{G}al(\mathbb{C}/\mathbb{Q})$-invariant.

This implies

Claim 1: Consider a subspace $W \subset V_\mathbb{C}$, and let $\tilde{W}_\mathbb{Q} \subset V_\mathbb{Q}$ be a smallest subspace of $V_\mathbb{Q}$ such that $\tilde{W}_\mathbb{Q} \otimes_\mathbb{Q} \mathbb{C} \supset W$. **Then $W_\mathbb{Q} \otimes_\mathbb{Q} \mathbb{C}$ is generated by $\sigma(W)$, for all $\sigma \in \mathcal{G}al(\mathbb{C}/\mathbb{Q})$.** ■

PROPOSITION: Let $V_\mathbb{Q}$ be a rational Hodge structure, ρ the corresponding $U(1)$-action, and MT its Mumford-Tate group. **Then MT is the Zariski closure of a group generated by $\sigma(\rho)$, for all $\sigma \in \mathcal{G}al(\mathbb{C}/\mathbb{Q})$.**

Proof: Since MT is rational, it is preserved by $\mathcal{G}al(\mathbb{C}/\mathbb{Q})$. The algebraic closure of the group generated by $\sigma(\rho)$ coincides with MT, because it is a smallest group which is rational, Zariski closed and contains ρ. ■

Transcendental Hodge lattice

THEOREM: Let $V_{\mathbb{Q}}$ be a rational, polarized Hodge structure of weight w, $V_C = \bigoplus_{\substack{p+q=w \\ p,q\geqslant 0}} V^{p,q}$, and $V^{tr} \subset V_{\mathbb{C}}$ a minimal rational subspace containing $V^{d,0}$. **Then V^{tr} is a Hodge substructure.**

Proof: By Claim 1, V^{tr} is generated by $\sigma(V^{d,0})$ for all $\sigma \in \mathcal{G}al(\mathbb{C}/\mathbb{Q})$. Since $V^{d,0}$ is preserved by the $U(1)$-action ρ, V^{tr} is preserved by the group generated by all $\sigma(\rho)$, which is MT. Finally, a rational subspace is a Hodge substructure if and only if it MT-invariant (Corollary 1). ∎

Polarization

DEFINITION: Polarization on a rational Hodge structrure of weight w is a $U(1)$-invariant non-degenerate 2-form $h \in V_{\mathbb{Q}}^* \otimes V_{\mathbb{Q}}^*$ (symmetric or antisymmetric depending on parity of w) which satisfies

$$-\sqrt{-1}^{\,p-q} h(x, \overline{x}) > 0 \qquad (*)$$

("Riemann-Hodge relations") for each non-zero $x \in V^{p,q}$.

EXAMPLE: Take the primitive part $HP^p(M) := \ker \Lambda|_{H^p(M)}$ of the cohomology of a projective manifold M, $\dim_{\mathbb{C}} M \geqslant 2p$. The **Riemann-Hodge form** $\eta, \eta' \to \int_M \eta \wedge \eta' \wedge \omega^{n-p}$, where $\omega \in H^{1,1}(M)$ is the fundamental class of a hyperplane section, **defines a polarization on $HP^p(M)$.**

DEFINITION: A **simple object** of an abelian category is an object which has no proper subobjects. An abelian category is **semisimple** if any object is a direct sum of simple objects.

CLAIM: Category of polarized Hodge structures in semisimple.

Proof: Orthogonal complement of a Hodge substructure $V' \subset V$ with respect to h is again a Hodge substructure, and this complement does not intersect V'; both assertions follow from the Riemann-Hodge relations. ■

Hyperkähler manifolds

DEFINITION: **Hyperkähler manifold** is a compact, holomorphically symplectic manifold M of Kähler type. We say that it has **maximal holonomy** or (IHS) if $\pi_1(M)=0$ and $H^{2,0}(M)=\mathbb{C}$. **We shall tacitly assume the maximal holonomy condition.**

REMARK: Calabi-Yau theorem implies that M **admits a Ricci-flat Kähler metric,** unique in each holonomy class. "Maximal holonomy" is maximal holonomy of the corresponding Levi-Civita connection.

REMARK: In differential-geometric terms, **maximal holonomy means that** $\mathcal{H}\mathrm{ol}(M)=Sp(n)$ **and not its strict subgroup**, where $\mathcal{H}\mathrm{ol}(M)$ is holonomy of Levi-Civita connection of the Calabi-Yau metric on a hyperkähler manifold M.

THEOREM: (Fujiki). Let $\eta\in H^2(M)$, and $\dim M=2n$, where M is hyperkähler. **Then $\int_M\eta^{2n}=cq(\eta,\eta)^n$, for some primitive integer quadratic form q on $H^2(M,\mathbb{Z})$,** and $c>0$ a rational number.

DEFINITION: The form q is called **Bogomolov-Beauville-Fujiki form**. It has signature $(3,b_2-3)$.

Hodge structures of K3 type

DEFINITION: A Hodge structure $V = V^{2,0} \oplus V^{1,1} \oplus V^{0,2}$ is called **of K3 type** if $\dim V^{2,0} = 1$. A **pseudo-polarization** on a Hodge structure of K3 type is a $U(1)$-invariant non-degenerate 2-form $q \in V_{\mathbb{Q}}^* \otimes V_{\mathbb{Q}}^*$ which is positive definite on $\operatorname{Re} V^{2,0}$ and has signature $(1, n)$ on $V^{1,1}$.

EXAMPLE: The Poincaré pairing q on $H^2(K3)$ **is a pseudo-polarization.**

EXAMPLE: Let M be a hyperkähler manifold, and q the Bogomolov-Beauville-Fujiki form on $H^2(M)$. **Then q is a pseudo-polarization**.

REMARK: A K3 surface (or a hyperkähler manifold) is projective **if and only if there exists a rational class $v \in H^{1,1}(M)$ such that $q(v, v) > 0$.**

Transcendental lattice in the Hodge structures of K3 type

PROPOSITION: Let V be a pseudo-polarized Hodge structure of K3 type, and $V_{tr} \subset V$ the transcendental lattice. **Then either:**

(a) There exists $v \in V_{\mathbb{Q}}^{1,1}$ such that $q(v,v) > 0$. In this case **the pseudo-polarization q restricted to V_{tr} is a polarization,** and V_{tr} is **irreducible as a Hodge structure.**

(b) The pseudo-polarization is negative definite on $V_{\mathbb{Q}}^{1,1}$. In this case V_{tr} **is also irreducible as a Hodge structure.**

(b) The lattice $V_{\mathbb{Q}}^{1,1}$ is negative semidefinite, and $V_{\mathbb{Q}}^{1,1} \cap (V_{\mathbb{Q}}^{1,1})^{\perp} = \langle\eta\rangle$. In this case, $V_{tr} \supset \langle\eta\rangle$, **hence it is not irreducible, but $\frac{V_{tr}}{\langle\eta\rangle}$ is an irreducible Hodge structure.**

Proof. Step 1: Let $W \subsetneq V_{tr}$ be a Hodge substructure. Since $W \neq V_{tr}$, $W \subset V^{1,1}$. Then $W^{\perp} \cap V_{tr}$ is a rational sublattice which contains $V^{2,0}$, hence $W^{\perp} \cap V_{tr} = V_{tr}$. This implies that $q|_W = 0$; since $V^{1,1}$ has signature $(1,n)$, this implies that $\dim W = 1$.

Step 2: In case q is non-degenerate on $V_{\mathbb{Q}}^{1,1}$, its orthogonal contains no rational (1,1)-vectors, hence V_{tr} is irreducible. If q is degenerate on $V_{\mathbb{Q}}^{1,1}$, it has signature $0,-,-,-,...-$, and $V_{tr}^{\perp} \cap V_{tr}$ is a vector of type (1,1) which generates $V_{\mathbb{Q}}^{1,1} \cap (V_{\mathbb{Q}}^{1,1})^{\perp}$. ■

Symplectic isogenies and Hodge structures

DEFINITION: Let A, B be holomorphically symplectic manifolds, and $Z \subset A \times B$ a correspondence. Consider the map $\eta \mapsto (\pi_2)_*(\pi_1^*\eta)$ induced by Z. We say that Z is **a symplectic isogeny** if $(\pi_2)_*(\pi_1^*\Omega) \neq 0$ for a holomorphically symplectic form Ω.

CLAIM: Let M_1, M_2 be K3 surfaces, and $Z \subset M_1 \times M_2$ a symplectic isogeny. **Then the map $\eta \mapsto (\pi_2)_*(\pi_1^*\eta)$ induces an isomorphism $T(M_1) \to T(M_2)$ of the (rational) transcendental Hodge lattice**.

Proof: Since $(\pi_2)_*\pi_1^*$ preserves rational classes and the Hodge types, it takes $T(M_1)$ to $T(M_2)$. This map is injective, because the only strict substructure of $T(M_i)$ is of rank 1 and type (1,1), and surjective for the same reason. ■

PROPOSITION: (Witt)
Let W be a non-degenerate rational quadratic lattice, and $W_1, W_2 \subset W$ isomorphic quadratic sublattices. **Then the orthogonal complements $W_1^\perp$, $W_2^\perp$ are isomorphic as quadratic lattices**.
Proof: Serre, Course of Arithmetics. ■

COROLLARY: Let M_1, M_2 be projective hyperkähler varieties which are symplectically isogeneous. **Then $H^2(M_1, \mathbb{Q})$ is isomorphic to $H^2(M_2, \mathbb{Q})$ as a rational Hodge structure.**
Proof: $T(M_1) = T(M_2)$ as shown above, and $T(M_1) + T(M_1)^\perp \cong T(M_2) + T(M_2)^\perp$ by Witt's theorem. ■

The “Shafarevich conjecture”

Torelli theorem: Let M_1, M_2 be K3 surfaces. **Then $M_1 \cong M_2$ if and only if their integer Hodge structures on H^2 are isomorphic.**

THEOREM: (N. Buskin)
Let M_1, M_2 be projective K3 surfaces with isomorphic rational Hodge structures on H^2. **Then M_1, M_2 are symplectically isogeneous.**

REMARK: This is more or less the Hodge conjecture for $M_1 \times M_2$.

“Shafarevich conjecture”: Let M_1, M_2 be hyperkähler manifolds in the same deformation class and with isomorphic rational Hodge structures on H^2. **Then M_1, M_2 are symplectically isogeneous.**

Why this is hard: To obtain an isogeny of K3 surfaces, we contract a bunch of rational curves and pass to an orbifold covering, obtaining a smooth K3 surface. Until recently, **there was no way to produce an isogeny of hyperkähler manifolds.**

Degenerate twistor deformations

DEFINITION: Let M be a smooth manifold. An **almost complex structure** is an operator $I: TM \longrightarrow TM$ which satisfies $I^2 = -\mathrm{Id}_{TM}$.

The eigenvalues of this operator are $\pm\sqrt{-1}$. The corresponding eigenvalue decomposition is denoted $TM = T^{0,1}M \oplus T^{1,0}(M)$.

DEFINITION: An almost complex structure is **integrable** if $\forall X, Y \in T^{1,0}M$, one has $[X,Y] \in T^{1,0}M$. In this case I is called **a complex structure operator**. A manifold with an integrable almost complex structure is called **a complex manifold**.

REMARK: The "usual definition": complex structure is an atlas on a manifold with differentials of all transition functions in $GL(n,\mathbb{C})$.

THEOREM: (Newlander-Nirenberg)
These two definitions are equivalent.

REMARK: An almost complex structure I **is uniquely determined by a subbundle $B \subset TM \otimes_{\mathbb{R}} \mathbb{C}$** such that $TM \otimes_{\mathbb{R}} \mathbb{C} = B \oplus \overline{B}$. Then we write $I = \sqrt{-1}$ on B and $I = -\sqrt{-1}$ on $\overline{B}$.

Null-space of a form and and Cartan's formula

DEFINITION: Let Ω be a differential form on M. The **kernel**, or **the null-space** $\ker(\Omega)\subset TM$ of Ω is the space of all vector fields $X\in TM$ such that the contraction $i_X(\Omega)$ vanishes.

Theorem 1: Let Ω be a differential p-form on M, $d\Omega=0$. **Then for any $X,Y\in\ker(\Omega)$, one has $[X,Y]\in\ker(\Omega)$.**

Proof. Step 1: Let $X,X_1\in\ker(\Omega)$, and $X_2,...,X_p$ any vector fields. Cartan's formula implies that $\operatorname{Lie}_X(\Omega)=d(i_X(\Omega))+i_X(d\Omega)=0$, hence $\operatorname{Lie}_X(\Omega)=0$.

Step 2: $\operatorname{Lie}_X(\Omega)(X_1,...,X_p)=\operatorname{Lie}_X(\Omega(X_1,...,X_p))-\sum_{i=1}^p\Omega(X_1,...,[X,X_i],...X_p)$. All terms of this sum, except $\Omega([X,X_1],X_2,...,X_p)$, vanish, because $X_1\in\ker(\Omega)$. Since $\operatorname{Lie}_X(\Omega)=0$, we have $\Omega([X,X_1],X_2,...,X_p)=0$ for all $X_2,...,X_p$. Therefore, $[X,X_1]\in\ker(\Omega)$. ∎

Corollary 1: Suppose that $\Omega\in\Lambda^2(M,\mathbb{C})$ be a closed p-form such that $\ker(\Omega)\cap T_{\mathbb{R}}M=0$ and $\ker(\Omega)\oplus\overline{\ker(\Omega)}=T_{\mathbb{C}}M$. Define $I:TM\longrightarrow TM$ by $I|_{\ker(\Omega)}=-\sqrt{-1}$ and $I|_{\overline{\ker(\Omega)}}=\sqrt{-1}$. **Then I defines a complex structure.**

Proof: I is by construction real and satisfies $I^2=-1$. The corresponding eigenspace $T^{1,0}M$ coincides with $\ker(\Omega)$, and Theorem 1 implies that I is integrable. ∎

Holomorphically symplectic manifolds

DEFINITION: Let (M,I) be a complex manifold, and $\Omega\in\Lambda^2(M,\mathbb{C})$ a differential form. We say that Ω is **non-degenerate** if $\ker\Omega\cap T_{\mathbb{R}}M=0$. We say that it is **holomorphically symplectic** if it is non-degenerate, $d\Omega=0$, and $\Omega(IX,Y)=\sqrt{-1}\,\Omega(X,Y)$.

REMARK: The equation $\Omega(IX,Y)=\sqrt{-1}\,\Omega(X,Y)$ means that **Ω is complex linear with respect to the complex structure on $T_{\mathbb{R}}M$ induced by I.**

REMARK: Consider the Hodge decomposition $T_{\mathbb{C}}M=T^{1,0}M\oplus T^{0,1}M$ (decomposition according to eigenvalues of I). Since $\Omega(IX,Y)=\sqrt{-1}\,\Omega(X,Y)$ and $I(Z)=-\sqrt{-1}\,Z$ for any $Z\in T^{0,1}(M)$, we have $\ker(\Omega)\supset T^{0,1}(M)$. Since $\ker\Omega\cap T_{\mathbb{R}}M=0$, real dimension of its kernel is at most $\dim_{\mathbb{R}}M$, giving $\dim_{\mathbb{R}}\ker\Omega=\dim M$. **Therefore,** $\ker(\Omega)=T^{0,1}M$**.**

COROLLARY: Let Ω be a holomorphically symplectic form on a complex manifold (M,I). **Then I is determined by Ω uniquely.**

Let's define complex structures in terms of complex-valued 2-forms!

Holomorphically symplectic forms and complex structures

CLAIM: Let M be a smooth $2n$-dimensional manifold. **Then there is a bijective correspondence between the set of almost complex structures, and the set of sub-bundles $T^{0,1}M \subset TM \otimes_{\mathbb{R}} \mathbb{C}$ satisfying $\dim_{\mathbb{C}} T^{0,1}M = n$ and $T^{0,1}M \cap TM = 0$** (the last condition means that there are no real vectors in $T^{1,0}M$, that is, that $T^{0,1}M \cap T^{1,0}M = 0$).

Proof: Set $I\big|_{T^{1,0}M} = \sqrt{-1}$ and $I\big|_{T^{0,1}M} = -\sqrt{-1}$. ■

Theorem 2: Let $\Omega \in \Lambda^2(M, \mathbb{C})$ be a smooth, complex-valued, non-degenerate 2-form on a $4n$-dimensional real manifold. Assume that $\Omega^{n+1} = 0$. Consider the bundle

$$T^{0,1}_\Omega(M) := \{v \in TM \otimes \mathbb{C} \ \ | \ \ \Omega \lrcorner v = 0\}.$$

Then $T^{0,1}_\Omega(M)$ satisfies assumptions of the claim above, hence **defines an almost complex structure I_Ω on M.** If, in addition, **Ω is closed, I_Ω is integrable.**

Proof: Rank of Ω is $2n$ because $\Omega^{n+1} = 0$ and it is non-degenerate. Then $\ker \Omega \oplus \overline{\ker \Omega} = T_{\mathbb{C}}M$. Now Theorem 2 follows from Theorem 1. ■

Holomorphic Lagrangian fibrations

DEFINITION: Let (M, Ω) be a holomorphically symplectic manifold, $\dim_{\mathbb{C}} M = 2n$. A complex subvariety $Z \subset M$ is called **holomorphically Lagrangian** if $\Omega|_Z = 0$ in all smooth points of Z and $\dim Z = n$.

DEFINITION: A holomorphic Lagrangian fibration is a holomorphic map $f: M \longrightarrow X$ with all fibers holomorphic Lagrangian.

REMARK: Nota bene: Neither X nor fibers of f need to be non-singular: the definition makes sense in singular situation.

DEFINITION: (1,1)-form on a complex manifold is a differential form which satisfies $\omega(IX, Y) = -\omega(X, IY)$ (same Hodge type as the Hermitian forms).

Matsushita Theorem

THEOREM: (Matsushita, 1997)
Let $\pi: M \longrightarrow X$ be a surjective holomorphic map from a hyperkähler manifold M to X, whith $0 < \dim X < \dim M$. Assume that $H^{2,0}(M) = \mathbb{C}$ and $\pi_1(M) = 0$. **Then $\dim X = 1/2 \dim M$, and the fibers of π are holomorphic Lagrangian** (this means that the symplectic form vanishes on $\pi^{-1}(x)$).

DEFINITION: Such a map is called **holomorphic Lagrangian fibration**.

REMARK: The base of π is conjectured to be $\mathbb{C}P^n$ if it is normal. Hwang (2007) proved that $X \cong \mathbb{C}P^n$, if it is smooth. Matsushita (2000) proved that it has the same rational cohomology as $\mathbb{C}P^n$.

Degenerate twistor deformations

THEOREM: Let (M,Ω) be a holomorphically symplectic manifold, $\dim_{\mathbb C} M = 2n$, and $f: M \longrightarrow X$ a holomorphic Lagrangian fibration. Consider a $(2,0)+(1,1)$-form $\eta\in \Lambda^2(X,\mathbb C)$. Then (a) $\Omega_\eta := \Omega + f^*\eta$ is a **non-degenerate form which satisfies $\Omega_\eta^{n+1}=0$, hence defines an almost complex structure.** (b) If, moreover, $d\eta=0$, this almost complex structure is integrable.

Proof: (b) follows from Theorem 2 and (a) immediately. Relation $\Omega_\eta^{n+1}=0$ is proving by writing Ω and η in a basis dp_i, dq_i such that $\Omega = \sum_i dp_i\wedge dq_i$, coordinates on X are p_i, and

$$\eta = \sum \beta_{ij} dp_i\wedge dp_j + \sum \alpha_i dp_i \wedge d\overline{p}_i,$$

where $\alpha_i, \beta_{ij}\in C^\infty X$. Non-degeneracy follows immediately, because (2,0)-part of Ω_η is $\Omega+\eta^{2,0}$, and it is non-degenerate. ■

DEFINITION: Let (M,Ω) be a holomorphically symplectic manifold, $\dim_{\mathbb C} M = 2n$, $f: M\longrightarrow X$ a holomorphic Lagrangian fibration, and $\eta\in\Lambda^{1,1}(X,\mathbb C)$ a closed (1,1)-form. Let $\Omega_{t\eta} := \Omega + t\eta$, where $t\in\mathbb C$, and let I_t be the complex structure associated with t. The family of complex structures I_t is called **a degenerate twistor deformation** of M.

Degenerate twistor deformations: their properties

1. If η is exact, this deformation is trivial by Moser's theorem.

2. When M is compact and of Kähler type, M is hyperkähler. In this case I_t is a limit of *twistor deformations.* Denote by $\Omega \subset \mathbb{C}$ the set of all t for which (M, I_t) admits a Kähler structure. **The manifolds (M, I_t) are projective for $t \in S$, where S is a dense, countable family $S \subset \Omega$, and non-algebraic for $t \notin S$.**

3. Each manifold (M, I_t) is equipped with a holomorphic Lagrangian fibration $f_t : (M, I_t) \longrightarrow X$. **The fibers and the base of f_t are isomorphic for all t.**

4. This deformation has an algebro-geometric interpretation **("Tate-Shafarevich deformation"): it is defined by a cocycle $H^1(X, \mathsf{Aut}_\pi(M))$,** where $\mathsf{Aut}_\pi(M)$ is the sheaf of fiberwise automorphisms of $\pi : M \to X$.

Degenerate twistor deformations and isogenies

THEOREM: Let $M_1, M_2 \stackrel{\pi}{:\to} X$ be two projective hyperkähler manifolds in the same degenerate twistor family. **Then there exists a finite covering $\tilde U$ of an open subset $U \subset X$ such that $M_1 \times_X \tilde U$ is isomorphic to $M_2 \times_X \tilde U$.**

Proof. Step 1: Let $M_0 \to U$ be a proper submersive abelian fibration, and $J(M_0)$ the sheaf of fiberwise parallel translations of M_0. We say that an abelian fibration $\pi : M' \to U$ **is a torsor over $J(M_0)$** if $J(M_0)$ acts on M' freely and transitively on fibers. Clearly, Then $H^1(X, J(M_0))$ classifies the torsors over $J(M_0)$. A torsor which has a section is trivial, that is, isomorphic to $J(M_0)$.

Suppose that the projections $M_i \times_X \tilde U \to \tilde U$ have sections $s_i : \tilde U \to M_i$. Then the corresponding torsors are trivial, hence $M_1 \times_X \tilde U$ is isomorphic to $M_2 \times_X \tilde U$. **It remains to produce a $\tilde U$ for which $M_i \times_X \tilde U \to \tilde U$ has a section.**

Step 2: A multisection is a subvariety $Z \subset M$ transversal to the fibers of π and projected to X surjectively. Clearly, for any projective M_i, the map $M_1, M_2 \stackrel{\pi}{:\to} X$ admits a multisection (just take general hyperplane of complimentary dimension to the fibers). After removing a closed subset of X, we may assume that the multisection takes $x \in U$ to precisely m points of the torus $\pi^{-1}(x)$. Taking Σ_m-covering of U, **we obtain a proper Lagrangian fibration which has a section.**

Step 3: Let $\tilde U_1$ and $\tilde U_2$ be coverings of open subsets of X such that M_i has a section over $\tilde U_i$. **Then M_i has a section over $\tilde U := \tilde U_1 \times_X \tilde U_2$.** ∎

REMARK: Let $\tilde X$ be the compactification of $\tilde U$ admitting a proper holomorphic map to X. Then $M_1 \times_X \tilde X$ is isomorphic to $M_2 \times_X \tilde X$ on an open subset of $\tilde X$, hence these varieties are bimeromorphic. The image of the graph of bimeromorphism in M **is a correspondence which defines a symplectic isogeny.**

Symplectic isogeny and Galois cohomology

THEOREM: Let $M_1, M_2 \stackrel{\pi}{:\to} X$ be two projective hyperkähler manifolds in the same degenerate twistor family. Consider the rational action of $J(M)$ on M_i, and let $J_r(M)$ denote the r-torsion. **Then there exists $r \in \mathbb{Z}^{>0}$ such that the quotients of $M_i/J_r(M)$ are isomorphic.**

Proof: The fibrations $M_1 \to X$ and $M_2 \to X$ define abelian varieties A_1, A_2 over the field $k(X)$ of rational functions on X. Let $J(A)$ denote the corresponding Jacobian, which is an abelian variety over $k(X)$ with a rational point; the varieties A_1, A_2 are torsors over $J(A)$, defined by classes $H^1(\mathcal{G}al_{k(X)}, J(A))$. By previous theorem, $A_1 \times_{x(X)} K \cong A_2 \times_{x(X)} K \cong J(A) \times_{x(X)} K$ for some finite extension $[K : k(X)]$. Therefore, these classes of these torsors belong to the image of the map $H^1(\mathcal{G}al(K : k(X)), J(A))$. Let r be the order of $\mathcal{G}al(K : k(X))$. Since the multiplication by r acts on $H^1(\mathcal{G}al(K : k(X)), J(A))$ as 0, this implies that the quotient of A_1 by r-torsion in $J(A)$ is equal to the quotient of A_2. ∎

The Eichler transvections

I AM NOT GOING TO SHOW THIS, IT'S NOT REALLY FINISHED YET

DEFINITION: Let (W,q) be a non-degenerate rational quadratic lattice, and $U\subset W$ a sublattice generated by x,y such that $q(x)=q(y)=0$ and $q(x,y)=1$. Consider the isometry of $x^\perp$ which takes $z\in x^\perp$ to $z+tq(y,z)x$, where $t\in\mathbb{Q}$. **Then this isometry is extended to a unique isometry of W**, which is written as $v\mapsto v-t(y,v)x+t(x,v)y-\frac{t^2}{2}(y,y)(x,v)x$

Proof: Lemma 3.1, *V. Gritsenko, K. Hulek and G.K. Sankaran Abelianisation of orthogonal groups and the fundamental group of modular varieties.* ■

REMARK: This isometry is called **the Eichler transvection**.

CLAIM: Let M_1, M_2 be two projective hyperkähler manifolds in the same degenerate twistor family, $T(M_1)$, $T(M_2)$ be their transcendental Hodge lattices, and $x\in H^2(M,\mathbb{Q})$ the pullback of the hyperplane section class on X, where $\pi: M_i\to X$ denotes the Lagrangian fibration. Then $T(M_2)=E(T(M_2))$, where E is an Eichler transvection obtained from x and some y.

Proof: $T(M_2)$ is the smallest rational lattice containing $\Omega+\lambda x$ and $T(M_2)$ is the smallest rational lattice containing Ω. They also satisfy $\langle T(M_1),x\rangle = \langle T(M_2),x\rangle$. Therefore, there is a transvection which takes one to another. ■

The proof of Shafarevich conjecture

THEOREM: Let M_1, M_2 be hyperkähler manifolds in the same deformation class and with isomorphic rational Hodge structures on H^2. **Then M_1, M_2 are symplectically isogeneous.**

Proof. Step 1: Let $\mathfrak{S}$ be the set of all rational vectors which are orthogonal to $T(M_1)$. Then $T(M_2)$ is obtained by taking an Eichler transvection E to $T(M_1)$ associated with some $x \in \mathfrak{S}$. Then we take $T(M_3)$ by taking Eichler transvection E to $T(M_1)$ associated with some $y \in E(\mathfrak{S})$, and so on. Therefore, to prove that all transcendental lattices are related by symplectic isogeny, **it suffices to show that the Eichler transvections associated with all $x \in \mathfrak{S}$ acts transitively on isomorphic transcendental Hodge lattices.**

Step 2: Let G be the group generated by Eichler transvections associated with all $x \in \mathfrak{S}$, and $U = T(M_1)$.